DO NOW — RECALL

Noughts & Crosses · Part 2 — Discover iteration

Name: _____ Class: _____ Date: _____

1. In a zero-indexed C# array, which element does array[2] access?

2. Name the three parts of a C# for loop header used to visit every array element.

3. In a C# 2D array organised as rows and columns, what does GetLength(0) return?

4. If a board has 3 rows and 3 columns, how many values does it contain?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The item at index 2 — the third item.

2. Model answer

A counter, condition and update.

3. Model answer

The number of rows.

4. Model answer

9.

By the end, you can…

• spot repeated row-printing instructions and replace them with a loop;

• explain why a loop inside another loop is useful for a 2D board;

• create a reusable DrawBoard() method using GetLength() .

Vocabulary — know these terms

Recall the term — 2D array

Recall the term — row

Recall the term — column

Recall the term — index

Recall the term — dimension

Recall the term — nested loop

Recall the term — GetLength()

Recall the term — traverse

NEW LEARNING

Read, annotate and explain

Yesterday: three similar instructions

```
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]);
```

The problem: repeated code is harder to maintain. Today we will discover why iteration makes the same board easier to draw.

Notice the repeated statements

Look at the three explicit row-printing instructions. The operation is the same; only the row index changes: 0, 1, 2 .

Replace the rows with one loop

The first loop repeats the same row-print statement. row changes automatically; each repetition chooses a different row.

Spot the second repetition

Inside each row, we still refer to columns 0, 1 and 2. A second loop can move across the columns of one row.

Put one loop inside the other

Outer loop = down the rows. Inner loop = across the columns. This is called nested iteration .

Make it reusable

GetLength(0) gives the row count; GetLength(1) gives the column count. Place the loops in DrawBoard() , then call it whenever the display needs refreshing.

Keep the board array as a class-level field so DrawBoard() can access it. Lesson 015 uses the same board for moves.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

From three statements to one loop, then two

Start with the old solution

```
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]);
```

It works. But it repeats the same pattern three times.

Replace the three row statements

```
for (int row = 0; row < board.GetLength(0); row++)
{
    Console.WriteLine(board[row,0] + " | " +
                      board[row,1] + " | " +
                      board[row,2]);
}
```

Keep the row output the same but allow row to change inside the loop.

Replace the repeated columns

```
for (int col = 0; col < board.GetLength(1); col++)
{
    Console.Write(board[0,col]);
    if (col < board.GetLength(1)-1)
        Console.Write(" | ");
}
Console.WriteLine();
```

Now each column is selected by col . After the third column, finish the row.

Combine both loops

```
for (int row = 0; row < board.GetLength(0); row++)
{
    for (int col = 0; col < board.GetLength(1); col++)
    {
        Console.Write(board[row,col]);
        if (col < board.GetLength(1)-1)
            Console.Write(" | ");
    }
    Console.WriteLine();
}
```

Predict how many times Console.Write(board[row,col]) executes: 9 .

Refine into DrawBoard()

```
public static void DrawBoard()
{
    for (int row = 0; row < board.GetLength(0); row++)
    {
        for (int col = 0; col < board.GetLength(1); col++)
        {
            Console.Write(board[row,col]);
            if (col < board.GetLength(1)-1)
                Console.Write(" | ");
        }
        Console.WriteLine();
    }
}
```

This method assumes board is stored at class level. Call DrawBoard() from Main() . You can reuse it after every move in Lesson 015.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Identify the pattern

The original problem stays visible as we improve the code.

```
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]);
```

Method / working:

2. Introduce a row loop

The original problem stays visible as we improve the code.

```
for (int row = 0; row < ?; row++)
```

Method / working:

3. Notice the columns

The original problem stays visible as we improve the code.

```
for (int col = 0; col < ?; col++)
```

Method / working:

4. Combine and explain

The original problem stays visible as we improve the code.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Start with what worked
Use your three row-print statements from Lesson 014. Run them unchanged.

Question 2

Replace repeated rows
Write one for loop that prints all three rows. Within the loop you may still access columns 0, 1 and 2 explicitly.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Replace repeated columns
Write a for loop to display one row, visiting columns using col . Print separators only between squares.

Question 4

Build DrawBoard()
Combine your two loops to visit all nine squares and put them inside DrawBoard() . Keep board accessible from the method and call it in Main() .

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

board.GetLength(1) returns the number of columns in the board.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.